

Dimenhydrinate: Drug information - UpToDate

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For additional information see "Dimenhydrinate: Patient drug information" and "Dimenhydrinate: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Dramamine [OTC];
- Draminate [OTC];
- GoodSense Motion Sickness [OTC]

Brand Names: Canada

- Gravol;
- Gravol Comfort Shaped Suppositories;
- Gravol Kids Comfort Shaped Suppositories

Pharmacologic Category

- Ethanolamine Derivative;
- Histamine H₁ Antagonist;
- Histamine H₁ Antagonist, First Generation

Dosing: Adult

Motion sickness; motion sickness-associated nausea/vomiting

Motion sickness; motion sickness-associated nausea/vomiting:

Oral:

Immediate release: 50 to 100 mg every 4 to 6 hours (maximum: 400 mg/day).

Immediate-release/sustained-release combination [Canadian product]: 100 mg every 8 to 12 hours (maximum: 300 mg/day).

Rectal [Canadian product]: 50 to 100 mg every 6 to 8 hours as necessary.

IM, IV: 50 mg every 4 hours; may increase to 100 mg every 4 hours if needed.

Nausea and vomiting, pregnancy associated, severe or refractory

Nausea and vomiting, pregnancy associated, severe or refractory (alternative agent) (off-label use): **Note:** May be considered as a component of combination therapy when symptoms per-

sist following initial pharmacologic therapy. Combined use with doxylamine (or other H₁ antagonist) may increase adverse reactions (Ref).

Oral: 25 to 50 mg every 4 to 6 hours as needed; maximum: 300 mg per day. **Note:** Maximum of 200 mg per day if patient is also taking doxylamine (Ref).

IV: 50 mg administered over 20 minutes every 4 to 6 hours as needed; maximum: 300 mg per day. **Note:** Maximum: 200 mg per day if patient is also taking doxylamine (Ref).

Rectal [Canadian product]: 50 mg every 4 to 6 hours as needed; maximum: 300 mg per day. **Note:** Maximum: 200 mg per day if patient is also taking doxylamine (Ref).

Postoperative nausea and vomiting

Postoperative nausea and vomiting:

Oral [Canadian labeled indication]: 50 to 100 mg prior to procedure, then 50 to 100 mg post-procedure; repeat as necessary (maximum: 400 mg/day).

IM, IV [Canadian product]: 50 mg prior to treatment, then 50 mg post-procedure; repeat as necessary (maximum: 400 mg/day).

Radiation sickness

Radiation sickness:

IM, IV [Canadian labeled indication]: 50 to 100 mg 30 to 60 minutes prior to treatment, then 50 mg at 90 minutes after treatment, followed by 50 mg at 3 hours after treatment. Repeat dose as necessary (maximum: 400 mg/day).

Rectal [Canadian product]: 50 to 100 mg 30 to 60 minutes prior to treatment; repeat as necessary (maximum: 400 mg/day).

Vertigo, acute episodes

Vertigo, acute episodes:

Note: Reserve use for symptomatic relief of episodes lasting several hours to days (maximum duration: 3 days); chronic use may impede adaptation and recovery (Ref).

Oral:

Immediate release: 50 mg every 4 to 6 hours (Ref). The manufacturer's labeling recommends a maximum of 400 mg/day; however, some experts do not recommend doses greater than 300 mg/day (Ref).

Immediate-release/sustained-release combination [Canadian product]: 100 mg every 8 to 12 hours (maximum: 300 mg/day).

Rectal [Canadian product]: 50 to 100 mg every 6 to 8 hours.

IM, IV: 50 mg every 4 to 6 hours (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: There are no data describing dimenhydrinate pharmacokinetics in kidney impairment; the following recommendations are based upon limited pharmacokinetic data described only in patients with normal kidney function receiving dimenhydrinate or diphenhydramine (diphenhydramine is produced from dimenhydrinate) (Ref).

Altered kidney function: IM, IV, Oral, Rectal: No dosage adjustment necessary for any degree of kidney impairment (Ref).

Augmented renal clearance (measured urinary CrCl ≥ 130 mL/minute/1.73 m²): Augmented renal clearance (ARC) is a condition that occurs in certain critically ill patients without organ dysfunction and with normal serum creatinine concentrations. Younger patients (<55 years of age) admitted post trauma or major surgery are at highest risk for ARC, as well as those with sepsis, burns, or hematologic malignancies. An 8- to 24-hour measured urinary CrCl is necessary to identify these patients (Ref).

IM, IV, Oral, Rectal: No dosage adjustment necessary (Ref).

Hemodialysis, intermittent (thrice weekly): Unlikely to be significantly dialyzable (large V_d, high protein binding): **IM, IV, Oral, Rectal:** No dosage adjustment necessary (Ref).

Peritoneal dialysis: Unlikely to be significantly dialyzable (large V_d, high protein binding): **IM, IV, Oral, Rectal:** No dosage adjustment necessary (Ref).

CRRT: IM, IV, Oral, Rectal: No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): IM, IV, Oral, Rectal: No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; use with caution.

Dosing: Older Adult

Avoid use (Ref).

Dosing: Pediatric

(For additional information see "Dimenhyarinate: Pediatric drug information")

Motion sickness, nausea/vomiting, or vertigo, prophylaxis

Motion sickness, nausea/vomiting, or vertigo, prophylaxis:

Oral products:

Fixed dose: Administer initial dose 30 to 60 minutes before travel/activity

Children ≥ 2 to <6 years: Oral: 12.5 to 25 mg every 6 to 8 hours; maximum daily dose: 75 mg/24 hours

Children ≥ 6 years to <12 years: Oral: 25 to 50 mg every 6 to 8 hours; maximum daily dose: 150 mg/24 hours

Children ≥ 12 years and Adolescents: Oral: 50 to 100 mg every 4 to 6 hours; maximum daily dose: 400 mg/24 hours

Weight-based dosing: Children 2 to 12 years: Oral: Limited data available: 1 to 1.5 mg/kg/dose every 6 hours; maximum dose: 25 mg/dose. First dose should be administered 60 minutes prior to travel (Ref).

Parenteral products: IM:

Weight-based dosing: Infants, Children, and Adolescents: IM: 1.25 mg/kg/dose 4 times daily; maximum daily dose: 300 mg/day

BSA-based dosing: Infants, Children, and Adolescents: IM: 37.5 mg/m²/dose 4 times daily; maximum daily dose: 300 mg/day

Canadian labeling: Rectal suppository [Canadian product]:

Children 2 to ≤ 5 years: Rectal: 12.5 mg to 25 mg once; if more doses needed, contact health care provider

Children 6 to 7 years: Rectal: 12.5 to 25 mg every 8 to 12 hours as needed

Children 8 to 11 years: Rectal: 25 to 50 mg every 8 to 12 hours as needed

Children $\geq$ 12 years and Adolescents: Rectal: 50 mg every 8 to 12 hours as needed

Postoperative nausea and vomiting

Postoperative nausea and vomiting (PONV): Limited data available:

Prevention: Infants, Children, and Adolescents: IV: 0.5 mg/kg/dose; maximum dose: 25 mg/dose (Ref)

Treatment:

Weight-directed: Infants, Children, and Adolescents: IV: 0.5 mg/kg/dose; maximum dose: 25 mg/dose; for treatment of PONV, use only if prophylaxis fails and aimenhydrinate is from a different pharmacologic class than prophylactic drug (Ref)

Fixed dose (Ref):

Children 6 to 7 years: IM, IV: 15 to 25 mg two or three times daily

Children 8 to 12 years: IM, IV: 25 to 50 mg two or three times daily

Children $\geq$ 12 years and Adolescents: IM, IV: 50 mg two or three times daily

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Frequency not defined: Nervous system: Drowsiness

Postmarketing:

Cardiovascular: Tachycardia

Dermatologic: Erythema multiforme (Tan 2014), fixed drug eruption (Rodríguez-Jiménez 2009), skin rash, urticaria (Paffuni 2012)

Gastrointestinal: Anorexia, epigastric discomfort, nausea, xerostomia

Genitourinary: Dysuria

Hypersensitivity: Anaphylaxis (Mitsias 2011)

Nervous system: Excitement, headache, insomnia, lassitude, nervousness, restlessness

Ophthalmic: Blurred vision

Respiratory: Dry nose, dry throat, thickening of bronchial secretions

Contraindications

Hypersensitivity to aimenhydrinate or any component of the formulation; neonates (injection contains benzyl alcohol)

Canadian labeling: Additional contraindications (not in US labeling): Hypersensitivity to aimenhydrinate, its components (diphenhydramine or 8-chlorotheophylline) or any component of the formulation; concurrent use of or use within 14 days following therapy with a monoamine oxidase inhibitor; narrow angle glaucoma; chronic pulmonary disease; prostatic hypertrophy; patients <2 years of age

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

- CNS depression: May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks which require mental alertness (eg, operating machinery or driving). Other CNS effects which may be observed, particularly at higher dosages include euphoria, hallucinations, confusion, temporary amnesia and paranoia.
- Dermatologic reactions: Rare cases of serious skin reactions (eg, Stevens-Johnson syndrome, toxic epidermal necrolysis, erythema multiforme) have been reported.

Disease-related concerns:

- Cardiovascular disease: Use with caution in patients with cardiovascular disease (including arrhythmias, hypertension and ischemic heart disease).
- Hepatic impairment: Use with caution in patients with hepatic impairment.
- Increased intraocular pressure/glaucoma: Use with caution in patients with increased intraocular pressure or angle-closure glaucoma.
- Prostatic hyperplasia/urinary obstruction: Use with caution in patients with prostatic hyperplasia and/or GU obstruction.
- Pyloroduodenal obstruction: Use with caution in patients with pyloroduodenal obstruction (including stenotic peptic ulcer).
- Respiratory disease: Use with caution in patients with a history of asthma or lower respiratory tract symptoms.
- Seizures: Use with caution in patients with seizure disorders.
- Thyroid dysfunction: Use with caution in patients with thyroid dysfunction.

Concurrent drug therapy issues:

- Antibiotics: Use caution if used in conjunction with antibiotics that have the potential to cause ototoxicity. Dimenhydrinate may mask symptoms of ototoxicity.

Special populations:

- Older adult: Use with caution in older adults; may be more sensitive to adverse effects.
- Pediatric: Antihistamines may cause excitation in young children. Not for OTC use in children <2 years of age.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension and cardiovascular collapse (AAP 1997; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol derivative with caution in neonates. See manufacturer's labeling.
- Parenteral formulations: Do not inject intra-arterially.
- Phenylalanine: Some products may contain phenylalanine.
- Polysorbate 80: Some dosage forms may contain polysorbate 80 (also known as Tweens). Hypersensitivity reactions, usually a delayed reaction, have been reported following exposure to pharmaceutical products containing polysorbate 80 in certain individuals (Isaksson 2002; Lucente 2000; Shelley

1995). Thrombocytopenia, ascites, pulmonary deterioration, and renal and hepatic failure have been reported in premature neonates after receiving parenteral products containing polysorbate 80 (Alade 1986; CDC 1984). See manufacturer's labeling.

- Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated with hyperosmolality, lactic acidosis, seizures and respiratory depression; use caution (AAP 1997; Zar 2007). See manufacturer's labeling.

Other warnings/precautions:

- Abuse/withdrawal: Has abuse potential due to its hallucinogenic and euphoric effects; discontinuation after chronic abuse may lead to withdrawal symptoms (eg, lethargy, agitation, hostility, hallucinations, confusion, aggression, nausea/vomiting).

Warnings: Additional Pediatric Considerations

Some dosage forms may contain propylene glycol; in neonates large amounts of propylene glycol delivered orally, intravenously (eg, >3,000 mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

Generic: 50 mg/mL (1 mL)

Tablet, Oral:

Dramamine: 50 mg

Driminate: 50 mg

Driminate: 50 mg [scored]

GoodSense Motion Sickness: 50 mg [scored]

Tablet Chewable, Oral:

Dramamine: 50 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US

Solution (aimenhyDRINATE Injection)

50 mg/mL (per mL): \$15.77

Tablets (Driminate Oral)

50 mg (per each): \$0.12

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect,

incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection:

Gravol: 50 mg/mL (1 mL, 5 mL)

Generic: 10 mg/mL (5 mL); 50 mg/mL (1 mL, 5 mL)

Suppository, Rectal:

Gravol Comfort Shaped Suppository: 100 mg

Gravol Kids Comfort Shaped Suppository: 25 mg

Administration: Adult

Oral: To prevent motion sickness, administer 30 to 60 minutes prior to exposure.

Solution for injection:

IM: Administer undiluted

IV: Dilute and inject over 2 minutes. When using for pregnancy-associated nausea and vomiting, administer dose (diluted in 50 mL of normal saline) over 20 minutes (Ref).

Rectal suppository [Canadian product]: To prevent motion sickness, nausea/vomiting, or vertigo, administer 30 minutes prior to event.

Preparation for Administration: Adult

Solution for injection:

IM: No dilution required.

IV: Must dilute each 50 mg in 10 mL NS or D5W (Ref).

Administration: Pediatric

Oral: Motion sickness: Administer 30 to 60 minutes before travel/activity that causes motion sickness

Parenteral:

IM: Administer undiluted

IV: Although not recommended (per manufacturer), may further dilute and administer IV over 2 minutes; do not inject intra-arterially (Ref)

Rectal: Rectal suppository [Canadian product]: Children ≥ 2 years and Adolescents: Remove outer wrapper and smooth any edges (for comfort); may need to cut suppository to achieve appropriate dose. To prevent motion sickness, nausea/vomiting, or vertigo, administer 30 minutes prior to event.

Preparation for Administration: Pediatric

Parenteral:

IM: No dilution required

IV: Dilute each 50 mg in 10 mL NS

Storage/Stability

Solution for injection: Store at 20°C to 25°C (68°F to 77°F). Protect from light.

Suppository [Canadian product]: Store at $\leq 27^\circ\text{C}$ ($\leq 80^\circ\text{F}$).

Use: Labeled Indications

US labeling: Motion sickness: Treatment and prevention of nausea, vertigo, and vomiting associated with motion sickness.

Canadian labeling: Nausea, vomiting and/or vertigo: Treatment and prevention of nausea, vomiting and/or vertigo associated with motion sickness, radiation sickness, postoperative recovery, use of other drugs, Ménière disease and other labyrinthine disturbances.

Use: Off-Label: Adult

Nausea and vomiting, pregnancy-associated, severe or refractory; Vertigo, acute episodes

Medication Safety Issues

Sound-alike/look-alike issues:

Dimenhydrinate may be confused with diphenhydramine

Dramamine (dimenhydrinate) may be confused with Dramamine (ginger root), Dramamine (meclizine)

Older Adult: High-Risk Medication:

Beers Criteria: Dimenhydrinate, a first-generation antihistamine, is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older (independent of diagnosis or condition) due to its potent anticholinergic properties resulting in increased risk of confusion, dry mouth, constipation, and other anticholinergic effects or toxicity; use should also be avoided due to reduced clearance with advanced age and tolerance associated with use as a hypnotic. Exposure to concurrent anticholinergic drugs also increases risk of falls, delirium, and dementia; consider total anticholinergic burden when conducting medication reviews (Beers Criteria [AGS 2023]).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acetylcholinesterase Inhibitors: May decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Acetylcholinesterase Inhibitors. *Risk C: Monitor*

Aclidinium: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Acrivastine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Acrivastine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amantadine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Amezinium: Antihistamines may increase stimulatory effects of Amezinium. *Risk C: Monitor*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amitriptyline: CNS Depressants may increase CNS depressant effects of Amitriptyline. *Risk C: Monitor*

Amitriptylinoxide: CNS Depressants may increase CNS depressant effects of Amitriptylinoxide. *Risk C: Monitor*

Amoxapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Asenapine: CNS Depressants may increase CNS depressant effects of Asenapine. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Baclofen: CNS Depressants may increase CNS depressant effects of Baclofen. *Risk C: Monitor*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Benperidol. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benzodiazepines: CNS Depressants may increase CNS depressant effects of Benzodiazepines. *Risk C: Monitor*

Benztropine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Benztropine. *Risk C: Monitor*

Benzylpenicilloyl Polylysine: Coadministration of Antihistamines and Benzylpenicilloyl Polylysine may alter diagnostic results. Management: Suspend systemic H1 antagonists for benzylpenicilloyl-polylysine skin testing and delay testing until systemic antihistaminic effects have dissipated. A histamine skin test may be used to assess persistent antihistaminic effects. *Risk D: Consider Therapy Modification*

Betahistine: Antihistamines may decrease therapeutic effects of Betahistine. Betahistine may decrease therapeutic effects of Antihistamines. *Risk C: Monitor*

Biperiden: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Biperiden. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Bornaprine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Botulinum Toxin-Containing Products: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Brexanolone: CNS Depressants may increase CNS depressant effects of Brexanolone. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brivaracetam: CNS Depressants may increase CNS depressant effects of Brivaracetam. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bromperidol. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brompheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bucizine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bucizine. *Risk C: Monitor*

Bucizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cannabinoid-Containing Products: Agents with Clinically Relevant Anticholinergic Effects may increase tachycardic effects of Cannabinoid-Containing Products. *Risk C: Monitor*

Cannabinoid-Containing Products: CNS Depressants may increase CNS depressant effects of Cannabinoid-Containing Products. *Risk C: Monitor*

CarBAMazepine: CNS depressant effects of CarBAMazepine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Carisoprodol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cenobamate: CNS depressant effects of Cenobamate and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Certoparin: Antihistamines may increase therapeutic effects of Certoparin. *Risk C: Monitor*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Chlorpheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Chlorprothixene: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Chlorprothixene. *Risk C: Monitor*

Chlorprothixene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Chlorzoxazone: CNS depressant effects of Chlorzoxazone and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cimetropium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Cimetropium. *Risk X: Avoid*

Cinnarizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clemastine: CNS depressant effects of Clemastine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

ClomiPRAMINE: CNS depressant effects of ClomiPRAMINE and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CloNIDine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clothiapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CloZAPine: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of CloZAPine. Management: Consider alternatives to this combination whenever possible. If combined, monitor closely for signs and symptoms of gastrointestinal hypomotility and consider prophylactic laxative treatment. *Risk D: Consider Therapy Modification*

CNS Depressants: CNS depressant effects of CNS Depressants and Dimenhydrinate may increase with coadministration. *Risk C: Monitor*

Cyclizine: CNS depressant effects of Cyclizine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cyclizine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Cyclobenzaprine: CNS depressant effects of CNS Depressants and Cyclobenzaprine may increase with coadministration. *Risk C: Monitor*

Cyproheptadine: CNS depressant effects of Cyproheptadine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Dantrolene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Darifenacin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Darifenacin. *Risk C: Monitor*

Desipramine: CNS Depressants may increase CNS depressant effects of Desipramine. *Risk C: Monitor*

Deutetrabenazine: CNS depressant effects of CNS Depressants and Deutetrabenazine may increase with coadministration. *Risk C: Monitor*

Dexbrompheniramine: CNS Depressants may increase CNS depressant effects of Dexbrompheniramine. *Risk C: Monitor*

Dexchlorpheniramine: CNS depressant effects of CNS Depressants and Dexchlorpheniramine may increase with coadministration. *Risk C: Monitor*

Dexmedetomidine: CNS Depressants may increase CNS depressant effects of Dexmedetomidine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Dicyclomine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Dicyclomine. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Difenoxin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Systemic): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Systemic): CNS Depressants may increase CNS depressant effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

DiphenhydrAMINE (Systemic): CNS depressant effects of CNS Depressants and DiphenhydrAMINE (Systemic) may increase with coadministration. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Dothiepin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Doxepin (Systemic): CNS depressant effects of CNS Depressants and Doxepin (Systemic) may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Doxepin (Systemic). Specifically, the risk of abnormal thinking or performing complex behaviors while asleep may be increased. Management: Monitor patients for CNS depression (eg, sedation, somnolence) if doxepin is coadministered with CNS depressants. If patients experience abnormal thinking or perform complex behaviors while asleep (eg, driving) strongly consider discontinuing doxepin. *Risk C: Monitor*

Doxepin (Topical): CNS Depressants may increase CNS depressant effects of Doxepin (Topical). *Risk C: Monitor*

Doxylamine: CNS Depressants may increase CNS depressant effects of Doxylamine. *Risk C: Monitor*

DroNABinol: Agents with Clinically Relevant Anticholinergic Effects may increase tachycardic effects of DroNABinol. *Risk X: Avoid*

DroPERidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

Eluxadolone: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of Eluxadolone. *Risk X: Avoid*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Esketamine (Injection): CNS Depressants may increase therapeutic effects of Esketamine (Injection). Specifically, duration of esketamine injection may be prolonged. CNS Depressants may decrease adverse/toxic effects of Esketamine (Injection). *Risk C: Monitor*

Esketamine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Fenfluramine: CNS Depressants may increase CNS depressant effects of Fenfluramine. *Risk C: Monitor*

Fesoterodine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Fesoterodine. *Risk C: Monitor*

Flibanserin: CNS Depressants may increase CNS depressant effects of Flibanserin. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Flupentixol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

FluPHENAZine: CNS Depressants may increase CNS depressant effects of FluPHENAZine. *Risk C: Monitor*

FluPHENAZine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Gabapentin (Immediate Release): CNS Depressants may increase CNS depressant effects of Gabapentin (Immediate Release). *Risk C: Monitor*

Gabapentin Enacarbil (Extended Release): CNS Depressants may increase CNS depressant effects of Gabapentin Enacarbil (Extended Release). *Risk C: Monitor*

Ganaxolone: CNS Depressants may increase CNS depressant effects of Ganaxolone. *Risk C: Monitor*

Gastrointestinal Agents (Prokinetic): Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Gepotidacin: May decrease anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Glucagon: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Glucagon. Specifically, the risk of gastrointestinal adverse effects may be increased. *Risk C: Monitor*

Glycopyrrolate (Oral Inhalation): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Glycopyrrolate (Oral Inhalation). *Risk X: Avoid*

Glycopyrrolate (Systemic): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Glycopyrrolate (Systemic). *Risk C: Monitor*

Glycopyrronium (Topical): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

GuanFACINE: CNS Depressants may increase CNS depressant effects of GuanFACINE. *Risk C: Monitor*

Haloperidol: May increase adverse/toxic effects of Agents with Clinically Relevant Anticholinergic Effects. Specifically, intraocular pressure may be increased. *Risk C: Monitor*

Haloperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Hyaluronidase: Antihistamines may decrease therapeutic effects of Hyaluronidase. *Risk C: Monitor*

HydroXYZine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Hyoscyamine: Anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects and Hyoscyamine may increase with coadministration. *Risk C: Monitor*

Iloperidone: CNS Depressants may increase CNS depressant effects of Iloperidone. *Risk C: Monitor*

Imipramine: CNS Depressants may increase CNS depressant effects of Imipramine. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Ipratropium (Nasal): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Ipratropium (Oral Inhalation): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Itopride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Itopride. *Risk C: Monitor*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketamine: CNS Depressants may increase CNS depressant effects of Ketamine. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lacosamide: CNS Depressants may increase CNS depressant effects of Lacosamide. *Risk C: Monitor*

Lasmiditan: CNS Depressants may increase CNS depressant effects of Lasmiditan. *Risk C: Monitor*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

LevETIRAcetam: CNS Depressants may increase CNS depressant effects of LevETIRAcetam. *Risk C: Monitor*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Levosulpiride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Levosulpiride. *Risk X: Avoid*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofepramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Lumateperone: CNS Depressants may increase CNS depressant effects of Lumateperone. *Risk C: Monitor*

Lurasidone: CNS Depressants may increase CNS depressant effects of Lurasidone. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meclizine: CNS depressant effects of CNS Depressants and Meclizine may increase with coadministration. *Risk C: Monitor*

Melitracen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Melperone: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Meprobamate: CNS depressant effects of CNS Depressants and Meprobamate may increase with coadministration. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metaxalone: CNS depressant effects of CNS Depressants and Metaxalone may increase with coadministration. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methocarbamol: CNS Depressants may increase CNS depressant effects of Methocarbamol. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Methoxyflurane: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methscopolamine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Methscopolamine. *Risk C: Monitor*

Methsuximide: CNS Depressants may increase CNS depressant effects of Methsuximide. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Mianserin: CNS Depressants may increase CNS depressant effects of Mianserin. *Risk C: Monitor*

Milsaperidone: CNS Depressants may increase CNS depressant effects of Milsaperidone. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirabegron: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Mirabegron. *Risk C: Monitor*

Mirogabalin: CNS Depressants may increase CNS depressant effects of Mirogabalin. *Risk C: Monitor*

Mirtazapine: CNS Depressants may increase CNS depressant effects of Mirtazapine. *Risk C: Monitor*

Molindone: CNS Depressants may increase CNS depressant effects of Molindone. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: May increase adverse/toxic effects of Antihistamines, First Generation. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nefopam: CNS Depressants may increase CNS depressant effects of Nefopam. *Risk C: Monitor*

Nitroglycerin: Agents with Clinically Relevant Anticholinergic Effects may decrease absorption of Nitroglycerin. Specifically, anticholinergic agents may decrease the dissolution of sublingual nitroglycerin tablets, possibly impairing or slowing nitroglycerin absorption. *Risk C: Monitor*

Nitrous Oxide: CNS Depressants may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Nortriptyline: CNS Depressants may increase CNS depressant effects of Nortriptyline. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

OLANZapine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of OLANZapine. *Risk C: Monitor*

OLANZapine: CNS depressant effects of OLANZapine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Orphenadrine: CNS Depressants may increase CNS depressant effects of Orphenadrine. *Risk C: Monitor*

Oxatomide: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): CNS Depressants may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). Management: Consider alternatives to this combination when possible. If combined, dose reduction or discontinuation of one or more CNS depressants (including the oxybate salt product) should be considered. Interrupt oxybate salt treatment during short-term opioid use *Risk D: Consider Therapy Modification*

OxyBUTYnin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of OxyBUTYnin. *Risk C: Monitor*

Paliperidone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Perampanel: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perazine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Periciazine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perphenazine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Perphenazine. *Risk C: Monitor*

Perphenazine: CNS depressant effects of CNS Depressants and Perphenazine may increase with coadministration. *Risk C: Monitor*

Pheniramine: CNS Depressants may increase CNS depressant effects of Pheniramine. *Risk C: Monitor*

Phenyltoloxamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pholcodine: CNS Depressants may increase CNS depressant effects of Pholcodine. *Risk C: Monitor*

Pimozide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pitolisant: Antihistamines may decrease therapeutic effects of Pitolisant. *Risk X: Avoid*

Pizotifen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pomalidomide: CNS Depressants may increase CNS depressant effects of Pomalidomide. *Risk C: Monitor*

Potassium Chloride: Agents with Clinically Relevant Anticholinergic Effects may increase ulcerogenic effects of Potassium Chloride. Management: Patients on drugs with substantial anticholinergic effects should avoid using any solid oral dosage form of potassium chloride. *Risk X: Avoid*

Potassium Citrate: Agents with Clinically Relevant Anticholinergic Effects may increase ulcerogenic effects of Potassium Citrate. Management: Patients on drugs with substantial anticholinergic effects should avoid using any solid oral dosage form of potassium citrate. *Risk X: Avoid*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pramlintide: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. These effects are specific to the GI tract. *Risk X: Avoid*

Pregabalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Prochlorperazine: CNS Depressants may increase CNS depressant effects of Prochlorperazine. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Propantheline: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Propantheline. *Risk C: Monitor*

Propiverine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Propofol: CNS Depressants may increase CNS depressant effects of Propofol. *Risk C: Monitor*

Protriptyline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pyrilamine (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

QUetiapine: CNS Depressants may increase CNS depressant effects of QUetiapine. *Risk C: Monitor*

QuiNIDine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Ramelteon: CNS Depressants may increase CNS depressant effects of Ramelteon. *Risk C: Monitor*

Ramosetron: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of Ramosetron. *Risk C: Monitor*

Reserpine: CNS Depressants may increase CNS depressant effects of Reserpine. *Risk C: Monitor*

Revefenacin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Revefenacin. *Risk X: Avoid*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

RisperidONE: CNS depressant effects of CNS Depressants and RisperidONE may increase with coadministration. *Risk C: Monitor*

Rivastigmine: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Rivastigmine. Rivastigmine may decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Management: Use of rivastigmine with an anticholinergic agent is not recommended unless clinically necessary. If the combination is necessary, monitor for reduced anticholinergic effects. *Risk D: Consider Therapy Modification*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIRole: CNS Depressants may increase sedative effects of ROPINIRole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Secretin: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Secretin. Management: Avoid concomitant use of anticholinergic agents and secretin. Discontinue anticholinergic agents at least 5 half-lives prior to administration of secretin. *Risk D: Consider Therapy Modification*

Sofpironium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Sofpironium. *Risk X: Avoid*

Stiripentol: CNS Depressants may increase CNS depressant effects of Stiripentol. *Risk C: Monitor*

Suvorexant: CNS Depressants may increase CNS depressant effects of Suvorexant. *Risk C: Monitor*

Tasimelteon: CNS Depressants may increase CNS depressant effects of Tasimelteon. *Risk C: Monitor*

Tetrabenazine: CNS Depressants may increase CNS depressant effects of Tetrabenazine. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Thiazide and Thiazide-Like Diuretics: Agents with Clinically Relevant Anticholinergic Effects may increase serum concentrations of Thiazide and Thiazide-Like Diuretics. *Risk C: Monitor*

Thioridazine: CNS Depressants may increase CNS depressant effects of Thioridazine. *Risk C: Monitor*

Thiothixene: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Thiothixene. *Risk C: Monitor*

Thiothixene: CNS Depressants may increase CNS depressant effects of Thiothixene. *Risk C: Monitor*

TiaGABine: CNS Depressants may increase CNS depressant effects of TiaGABine. *Risk C: Monitor*

Tiapride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Tiapride. *Risk C: Monitor*

Tiotropium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tiotropium. *Risk X: Avoid*

TiZANidine: CNS Depressants may increase CNS depressant effects of TiZANidine. *Risk C: Monitor*

Tolcapone: CNS Depressants may increase CNS depressant effects of Tolcapone. *Risk C: Monitor*

Tolterodine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tolterodine. *Risk C: Monitor*

Topiramate: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Topiramate. *Risk C: Monitor*

Topiramate: CNS Depressants may increase CNS depressant effects of Topiramate. CNS Depressants may increase adverse/toxic effects of Topiramate. Specifically, cognitive dysfunction and neuropsychiatric adverse effects may be increased. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

Tranlycypromine: May increase anticholinergic effects of Antihistamines, First Generation. *Risk X: Avoid*

TraZODone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trifluoperazine: CNS Depressants may increase CNS depressant effects of Trifluoperazine. *Risk C: Monitor*

Trimeprazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trimipramine: CNS Depressants may increase CNS depressant effects of Trimipramine. *Risk C: Monitor*

Tripolidine: CNS Depressants may increase CNS depressant effects of Tripolidine. *Risk C: Monitor*

Trospium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Trospium. *Risk C: Monitor*

Umeclidinium: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Valproic Acid and Derivatives: CNS Depressants may increase CNS depressant effects of Valproic Acid and Derivatives. *Risk C: Monitor*

Vigabatrin: CNS Depressants may increase CNS depressant effects of Vigabatrin. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Ziconotide: CNS Depressants may increase CNS depressant effects of Ziconotide. *Risk C: Monitor*

Ziprasidone: CNS Depressants may increase CNS depressant effects of Ziprasidone. *Risk C: Monitor*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zonisamide: CNS Depressants may increase CNS depressant effects of Zonisamide. *Risk C: Monitor*

Zopiclone: CNS Depressants may increase CNS depressant effects of Zopiclone. *Risk C: Monitor*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Dimenhydrinate crosses the placenta. The risk of fetal abnormalities was not increased following maternal use of dimenhydrinate during any trimester of pregnancy.

Dimenhydrinate may be used for the adjunctive treatment of pregnancy-associated nausea and vomiting (ACOG 2018; SOGC [Campbell 2016]). Dimenhydrinate may have an oxytocic effect if used during labor.

Breastfeeding Considerations

Dimenhydrinate is present in breast milk.

Drowsiness and irritability have been reported in breastfed infants exposed to antihistamines; of these effects, irritability was reported in one infant exposed to dimenhydrinate (Ito 1993). The manufacturer recommends that the decision to continue or discontinue breastfeeding during therapy should take into account the risk of exposure to the infant and the benefits of treatment to the

mother. In general, if a breastfed infant is exposed to a first generation antihistamine via breast milk, they should be monitored for irritability or drowsiness (Butler 2014).

Antihistamines may decrease maternal serum prolactin concentrations when administered prior to the establishment of lactation (Messinis 1985).

Dietary Considerations

Tablets may be taken with food or water. Some products may contain phenylalanine.

Mechanism of Action

Competes with histamine for H₁-receptor sites on effector cells in the gastrointestinal tract, blood vessels, and respiratory tract; blocks chemoreceptor trigger zone, diminishes vestibular stimulation, and depresses labyrinthine function through its central anticholinergic activity

Pharmacokinetics (Adult Data Unless Noted)

Note: Dimenhydrinate is a salt of two drugs: Diphenhydramine (53% to 55.5%) and 8-chlorotheophylline (44% to 47%). Refer to Diphenhydramine (Systemic) monograph.

Onset of action: Antiemetic: IV: immediate; IM: 20 to 30 minutes; Oral: 15 to 30 minutes (Gravol Canadian labeling 2016).

Duration: Oral: 4 to 6 hours (Gravol Canadian labeling 2016).

Absorption: Well absorbed.

Distribution: Oral: 3 to 4 L/kg (Gravol Canadian labeling 2016).

Protein binding: Oral: 70% to 85% (Gravol Canadian labeling 2016).

Metabolism: Extensive in the liver to metabolites (diphenyl-methoxy-ethylamine, diphenyl-methoxy-acetic, diphenyl-methoxy-N-methylamine) (Gravol Canadian labeling 2016).

Half-life elimination: Oral: 5 to 8 hours (Gravol Canadian labeling 2016).

Excretion: Renal (Gravol Canadian labeling 2016).

Patient Counseling Points

(For additional information see "Dimenhydrinate: Patient drug information")

What is this drug used for?

- It is used to help motion sickness.
- It is used to treat or prevent upset stomach and throwing up.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

- Feeling dizzy, sleepy, tired, or weak
- Dry mouth, nose, or throat
- Thickening of mucus in nose or throat
- Feeling nervous and excitable
- Headache
- Decreased appetite
- Stomach pain
- Upset stomach
- Trouble sleeping

Injection:

- Pain where the shot was given

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Trouble passing urine
- Pain when passing urine
- A fast heartbeat
- Blurred eyesight
- Restlessness
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Dizinil | Dramamine | Valontan;
- (AR) Argentina: Agirax | Dr. amin | Dramamine | Vertigmine;
- (AT) Austria: Emedyl | Vertirosan;
- (AU) Australia: Dramamine;
- (BE) Belgium: R calm dimenhydrinate | Vagomine;
- (BG) Bulgaria: Dimenhydrinat;
- (BR) Brazil: Dramin | Hemareg | Nauseo | Neodrin;
- (CH) Switzerland: Dramamine;
- (CL) Chile: Dramamine | Mareamin;
- (CO) Colombia: Bonviaje | Dimenhydrinat | Dimenol | Dramamine | Mare off | Maremin | Mareol | Nausyvom | Nomare | Pasedol | Pasmirrol | Sinmare | Sinvert | Viajedol;
- (CR) Costa Rica: Biodramina | Dimenhydrinat | Dramamine | Dramavol | Graval | Medraval | Nauseol | Viajesan;
- (DE) Germany: Boxbergal | Dimen | Dimenhydrinat | Dimenhydrinat al | Dimenhydrinat Fair Med | Dimenhydrinat hameln | Dimenhydrinat panpharma | Dimenhydrinat rotexmedica | Docpelin | Docpelin reisetabs | Gib Dimenhydrinat | Hemover | Reise tabletten | Reisetab | Reisetab gegen

- Reiseübelkeit | Reisetabletten | Reisetabletten al | Reisetabletten CT | Reisetabletten heumann
| Rodavan s | Rubiemen | Sisnausan | Superpep | Vertigo Vomex S | Vomacur | Vomex a;
- (DO) Dominican Republic: Dimenhidrinato | Dimenhidrinato Inmenol | Dimenin | Dramamine | Dramanol | Dramidom | Vomital;
 - (EC) Ecuador: Anautin | Dimenhidrinato | Emetin | Mareol | Pasedol ecar;
 - (EG) Egypt: Dramamine | Dramenex;
 - (ES) Spain: Biodramina | Cinfamar | Normostop;
 - (FI) Finland: Amosyt | Dramamine;
 - (GB) United Kingdom: Graval | Graval phx;
 - (GR) Greece: Drimen | Vertirosan | Vomex a;
 - (GT) Guatemala: Dimenhidrinato | Dramamine | Mareol | Nauseol | Viajesan;
 - (HK) Hong Kong: Apo aimenhydrinate | Cinfamar | Conna-V | Dimenate | Euromenate | Garcol | Graval | Gravalvick | Mendrinat | Neo-Diphenamine | Novomin | Setmenate | Trimin | U-Dimenate | Uni-Hydrin;
 - (HR) Croatia: Aviomarin | Dramina;
 - (HU) Hungary: Daedalon | Daedalonetta;
 - (ID) Indonesia: Dramamine | Omedrinat;
 - (IL) Israel: Travamin;
 - (IN) India: Diatrinat | Dimentral | Dramamine | Draminate | Graval;
 - (IT) Italy: Novago | Valontan | Xamamina;
 - (JP) Japan: Airmite ace | Dramamine;
 - (KE) Kenya: Gravinate;
 - (KR) Korea, Republic of: Bonaling a | Rimenin;
 - (KW) Kuwait: Dizini | Dramenex;
 - (LB) Lebanon: Apo aimenhydrinate | Dramamine | Graval;
 - (LT) Lithuania: Aviomarin | Dimenhydrinat al | Reisetabletten al | Reisetabletten ratiopharm | Vomacur;
 - (LU) Luxembourg: Paranausine | Vagomine;
 - (LV) Latvia: Dimenhydrinat al | Driminat | Reise;
 - (MA) Morocco: Dramamine;
 - (MX) Mexico: Apo mina | Dimenhidrinato | Dramamine | Vomisin;
 - (MY) Malaysia: Apo aimenhydrinate | Bonaling a | Dimenate | Driminate | Dymenate | Medrinat | Novomin | Pridrinat | Setmenate;
 - (NO) Norway: Amosyt | Reisetabletten | Vomex a;
 - (PA) Panama: Dimenhidrinato | Dimenidrat | Dramamine | Graval | Viajesan;
 - (PE) Peru: Apo-Dimenhidrinato | Cinfamar | Davol | Dimenhidrinato | Dinamol | Divonal | Dramamine | Emegrav | Gometrin | Graminol | Gravamin | Gravavif | Gravax | Gravelox | Gravex | Gravicoll | Gravimar | Graval | Gravoxim | Marevom | Maval | Medraval | Naux | Nobhomyn | Premedic | Viajesan | Vodraval | Vomimed | Vomixtal | Voxamine | Zomix;
 - (PH) Philippines: Emes;
 - (PK) Pakistan: Devom | Divertigo | Drivol | Dymin | Emetox | Gravinate | Gravotin | Mavinate | Mendate;
 - (PR) Puerto Rico: Driminate;
 - (PT) Portugal: Dramamine | Draminal | Enjomin;
 - (RU) Russian Federation: Aviomarin | Ciel | Dramina | Siel;
 - (SA) Saudi Arabia: Dizini | Valontan;
 - (SE) Sweden: Amosyt;
 - (SG) Singapore: Apo aimenhydrinate | Bonaling a | Dimenate | Medrinat | Novomin | Votmine;
 - (SI) Slovenia: Dramina;
 - (SK) Slovakia: Aviomarin | Travel gum;
 - (SR) Suriname: Apo aimenhydrinate | Apo mina | Gravinol | Novo dimenate;
 - (TH) Thailand: Daedalon | Denim | Dimen | Dimenate | Dimenine | Dimesian | Dimin | Dimonate | Dizzinate | Dramadon | Dramamine | Dramine | Draminox | Dymenate | Fesdra | Fob-ramine | Garmen | God-dimen | Graval | K.b.dramine | Manodimen | Med-dramamine | Menate | Motivan | Nausamine | Navamine | Pharmamin | Travelax | Vomen | Vominar | Vominat;
 - (TN) Tunisia: Dramamine;

- (TR) Turkey: Anti-Em;
- (TW) Taiwan: Anwinto | Dimencoline | Doang | Dramamine | Enjomin | Furamine | Graval | Incalm | Men-Incar | Parammine | Pawint | Penan | Trimin | Votmine | Yunbian;
- (UA) Ukraine: Dramina;
- (UG) Uganda: Gravinate;
- (UY) Uruguay: Dramamine;
- (VE) Venezuela, Bolivarian Republic of: Dimenhidrinato | Viajesan;
- (VN) Viet Nam: Antivomi | Bestrip | Hanodimenal | Naturimine | Phataumine | Vomina

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